

Bank Management System

A comprehensive web application enabling banks to manage customer accounts, employees, and banking operations with role-based access control, email verification, and loan management.

Frontend

Angular with reactive forms and route guards

Backend

Spring Boot with JWT authentication

Database

PostgreSQL / MySQL



Email Verification

Secure registration with unique verification links and token expiry



Role-Based Access

Customer, Employee, and Admin roles with restricted endpoints



Loan Management

Application submission, approval workflow, and status notifications



Account Operations

Balance checks, transaction history, and account lifecycle management

Core User Roles



Customer

- Create and manage accounts
- Apply for loans
- View profile and transactions



Bank Employee

- Manage customer accounts
- Approve/reject loan applications
- Generate reports



Admin

- Manage employees
- Configure system settings
- Oversee operations

Database Architecture



Users

Authentication, roles, verification status



Accounts

Account numbers, balances, status tracking



Loans

Applications, amounts, approval workflow



Tokens

Verification links with expiry

Security Implementation

01

JWT Authentication

Secure API access with role claims in tokens

02

Token Validation

JwtAuthenticationFilter validates all requests

03

User Loading

UserDetailsService loads users by email

04

Endpoint Protection

@PreAuthorize annotations restrict access by role

Future Enhancements

Payment Gateway Integration

Enable seamless online transactions and transfers

Multi-Factor Authentication

Enhanced security with MFA implementation

Microservices Architecture

Scalable system design for growing demands

Analytics Dashboard

Comprehensive reporting and business intelligence